

# Comparing Web Browsing Behaviors with High and Low Information Literacy: A Case Study for Fact Check Against GPT Generated Fake News

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## ABSTRACT

Generative AI has introduced significant concerns about how people interact with information in society, particularly regarding the potential harm caused by fake news. To address this issue, it is critical to understand how people perceive fake news and how their information literacy can be improved. Our research tackles two key questions: "Can we estimate the level of information literacy from participants' web browsing behavior?", and "What characteristics can be observed when comparing participants with high and low information literacy?". We recruited 20 university students to read and evaluate the veracity of generated news. During this process, participants used web searches to verify the news, with their browsing logs collected using the open-source browser extension *TrackThink Camera*. The study achieved a 73.8 % accuracy in estimating participants' ability to detect fake news based on their web browsing behavior. Notably, we observed that web search time ratio is higher, the number of searches are more, and the average scrolling speed is faster for high information literacy participants.

## CCS CONCEPTS

• **Human-centered computing** → *Social media; Empirical studies in HCI.*

## KEYWORDS

Fake news, information literacy, web browsing, machine learning

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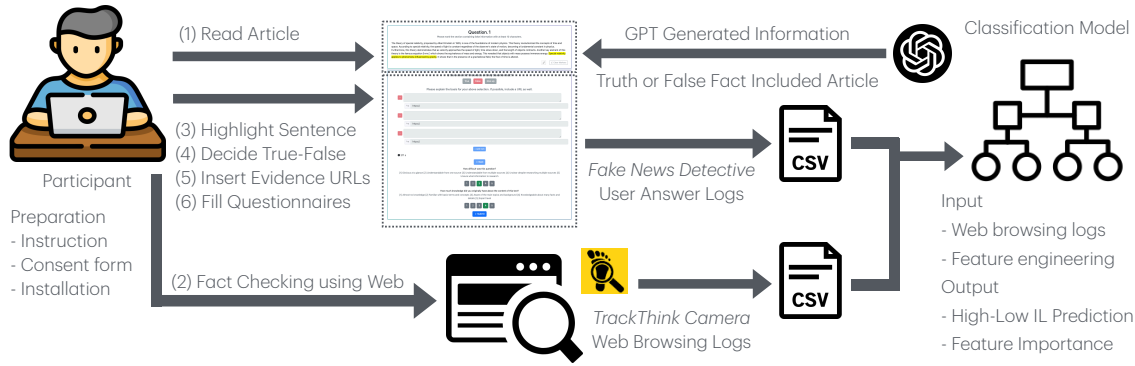
## 1 INTRODUCTION

Generative AI, as a relatively new technology, has been used to generate fake news to for the purpose of deceiving people [2, 9, 19]. In addition, the popularity and use of social media has acted as a facilitator in spreading fake news. A 2008 survey observed that social media is perceived as a form of news media for Americans under 30 [16]. Posting of media, especially by influencers, makes it easy for people to trust the information as factual news [8]. In addition, unwittingly receiving misinformation can also lead individuals to re-spreading the contents [15], which profoundly impacts society. To address this problem, it is important to develop methods that help to prevent humans from being deceived by false information.

There are various research efforts that aim to prevent deception by fake news. "Fake news detection" is one of the key concepts in this field [18]. Conroy et al. presents work on detecting false information using a linguistic cue and network analysis approach [3]. The work shows the potential to extract inappropriate information involved in the context by structuring information. Porto-Capetillo et al. proposed classifying input new articles into true, partially false, false, or other [14]. The best result was an F1 score of 0.295, using stylometric characteristics such as the number of uppercase characters, repeated sentences, symbols, and spelling errors. These studies aim to recognize false information from the text and filter it from being public. While filtering is an essential approach against the spread of fake news, improving people's information literacy is also a goal.

According to Eisenberg, information literacy is a basic skill required for life in the 21 century [5]. One approach for improving literacy is to enhance evaluation skills [4]. Eagleton and Dobler stated that evaluation skills are essential while reading, listening, and viewing. To "Weigh information carefully and wisely to determine its quality" [13] is a significant concept that guides this work. Our study focuses on this direction to support humans in improving their information literacy.

This study aims to explore how humans use web search behaviors while fact-checking. The primary objective of this study is to compare the characteristics of individuals with high and low ability to discern true and false information. By doing so, we seek to provide insights that can significantly enhance information literacy,



**Figure 1: Study overview: Participants use *Fake News Detective* and *TrackThink Camera* [20] in order to collect web browsing logs while fact-checking, truth or false decision toward GPT generated articles, evidence url against the decision, and subjective answers on questionnaires. Collected dataset is used to classify and compare characteristic differences between high and low information literacy (Explained as “IL” in the figure) participants.**

a skill of paramount importance in the digital age. To do so, we focus on answering two research questions (RQ):

- RQ1 Can we classify high and low information literacy participants’ from web browsing behaviors?
- RQ2 What characteristics can be observed when comparing participants with high and low information literacy?

## 2 RELATED WORK

In this section, we first discuss approaches to generating fake news and then present research on critical online reasoning.

### 2.1 Generation of Fake News

Huang et al. proposed a method for generating fake news containing false information with a style of mimicking actual news using natural language processing technologies such as BERT and RoBERTa [6]. They began by selecting parts of actual news articles collected from reliable sources and inserting false information. Additionally, they enhanced the persuasiveness of the articles by incorporating propaganda techniques. They successfully created high-quality fake news that is difficult to distinguish from fact-based news. As a result, the generated fake news proved highly effective in training fake news detection models.

### 2.2 Critical Online Reasoning

Recent studies have focused on critical online reasoning (COR) to assess the ability to analyze online information critically. Schmidt et al. conducted experiments using a new assessment tool called CORA, and eye-tracking to evaluate the COR of college students [17]. Thirty-two students participated in tasks assessing the credibility of websites and tweets, and their processes were analyzed. The results identified two distinct information processing patterns among college students: strategic and avoidant. Strategic processors efficiently processed information and scored higher on tasks. Nagel et al. analyzed the impact of web search behaviors and website characteristics on the critical reasoning of college students online [12]. They conducted tasks where college students performed

web searches to identify reliable sources on specific topics. The analysis revealed that web search behaviors and the quality of visited websites positively influenced students’ critical thinking skills.

## 3 METHODOLOGY

This section explains how we generate true and false news articles using large language model, followed by the data collection procedure. Lastly, we explain the approach used in the data analysis.

### 3.1 True and False Article Creation by GPT-4

In this study, we use GPT-4 [1] for generating true and false articles. This process uses prompts to create articles on specific topics.

In our prompt for article generation, we require following conditions: (1) The length of the article should be approximately 400 characters. (2) An article with false information must contain only one fictional element. (3) The fictional element must directly contradict a factual statement. (4) Avoid subjective phrases such as “surprisingly” or “interestingly” in the composition of the articles.

In the finalizing section, we modify the generated article if it does not follow the instructions. Also, we conduct fact-checking on each generated article. Web browsing is used to verify the facts in each sentence of the article and confirm the validity of the issues.

### 3.2 Data Collection Application

We use *Fake News Detective* and *TrackThink Camera* in this study.

**3.2.1 Fake News Detective.** We implement an application called *Fake News Detective*<sup>1</sup> for collecting participant’s question-solving results. This application shows generated articles explained in Section 3.1, selection button for *True*, *False*, or *Give up* decision, and subjective questionnaire feedback. This application collects logs of the start and stop timestamp while question-solving, true-or-false decision answers, highlighted sentences, generated articles, evidence URLs, and numeric feedback for the subjective questionnaire.

**3.2.2 TrackThink Camera.** We use an open-source web browsing extension called *TrackThink Camera* [20] to collect web browsing

<sup>1</sup><https://fake-news-quiz.vercel.app/>

**Table 1: Features engineering from web browsing logs collected by *TrackThink Camera (TTC)* and participant answers collected by *Fake News Detective (FND)*.**

| Feature                      | Description                                                   | Formulation                                                     |
|------------------------------|---------------------------------------------------------------|-----------------------------------------------------------------|
| time_from_start_to_search    | Time from starting the problem to beginning the search.       | $Timestamp\_TTC_{start} - Timestamp\_FND_{start}$               |
| time_from_start_to_end       | Time from starting the problem to submitting the answer.      | $Timestamp\_FND_{end} - Timestamp\_FND_{start}$                 |
| search_time_ratio            | Proportion of time spent searching while solving the problem. | $Time\_Duration\_Search / Time\_Duration\_All$                  |
| number_of_visited_sites      | Number of sites visited.                                      | $COUNT(URL_{all})$                                              |
| repeat_visit_count           | Number of times the same site was visited repeatedly.         | $COUNT(URL_{duplicate})$                                        |
| number_of_returned_questions | Number of times returned to the question page.                | $COUNT(URL_{FND})$                                              |
| number_of_closes             | Number of times tabs were closed.                             | $COUNT(Webpage\_Tab\_Close)$                                    |
| number_of_searches           | Number of times a search was conducted.                       | $COUNT(Query_{all})$                                            |
| query_noun_ratio             | Proportion of nouns in the search query.                      | $COUNT(Query_{noun}) / COUNT(Query_{all})$                      |
| average_search_query_length  | Average query length of search queries.                       | $\frac{1}{N} \sum_{i=1}^N query\_length_i$                      |
| average_scroll_speed         | Average speed of scrolling.                                   | $\mu = \frac{1}{N} \sum_{i=1}^N scroll\_speed_i$                |
| variance_of_scroll_speed     | Variance of scroll speed.                                     | $\sigma^2 = \frac{1}{N} \sum_{i=1}^N (scroll\_speed_i - \mu)^2$ |

logs. The application was integrated from *TrackThinkTS* [10, 11], which collects web browsing logs such as information about the visited website. The logs include URL, access page title, access page content, timestamp, clipboard copy, scroll speed, and scroll amount.

### 3.3 Participants

We recruited 20 (17 male and three female) participants in this study with an age range between 21 and 25 ( $M=22.6$ ). All participants were Japanese, and all were university or graduate students. Participants were allowed to opt out of the experiment at any time. Participants were asked to use the Google Chrome browser<sup>2</sup> in default. At the end of the experiment, all participants who completed it received a 3000 yen Amazon voucher. The study was conducted with approval from the Osaka Metropolitan University Graduate School of Informatics ethics committee.

### 3.4 Data Collection Procedure

Figure 1 shows the overall study flow. First participants access the *Fake News Detective* interface and start reading the generated article. Participants will use web browsing to fact-check the article, excluding GPTs, ensuring they are thorough and accurate in their research. Then, the participant highlights the sentence if they find false information. When the participant found evidence URLs, insert the URL into the application. Once the participant is satisfied with fact-checking, decide one option from *True*, *False*, or *Give up* button. The above process, including fact-checking and answering subjective questionnaires, is designed to be completed within a time limit of five minutes, encouraging participants to be efficient in their fact-checking. Finally, participants answered subjective questionnaires about “How difficult was this question?” and “How much knowledge did you originally have about the content of this text?” Participants conduct 18 (nine true and nine false) articles in total.

## 4 DATA ANALYSIS

In this section, we explain how we apply feature engineering and define high and low information literacy participants.

<sup>2</sup><https://www.google.com/intl/ja/chrome/>

### 4.1 Feature Engineering

To evaluate the dataset, user answer logs collected by *Fake News Detective* and web browsing logs collected by *TrackThink Camera* are synchronized using their respective timestamps. After concatenating two datasets, we apply feature engineering. Table 1 show list of features extracted in this study.

### 4.2 High-Low Information Literacy Definition

High and low information literacy participant is evaluated using ground truth data, participant decisions, and highlighted sentences. In this study, we used the Jaccard coefficient [7] to evaluate whether participants appropriately made decision against articles. When evaluators deem *fake news* articles, they highlight the incorrect sentence. The degree of match between the highlighted and expected sentence is measured using the Jaccard coefficient. The Jaccard coefficient is calculated using the following equation (1):

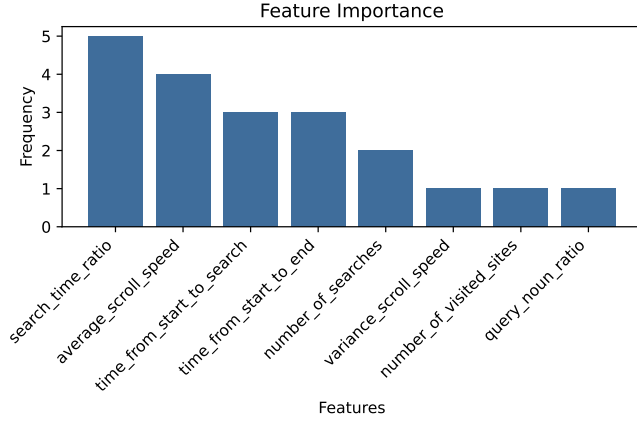
$$J(A, B) = \frac{|A \cap B|}{|A \cup B|} \quad (1)$$

Where  $A$  is the set of expected false word positional indexes, and  $B$  is the set of word positional indexes highlighted by the participant. The value of *MatchScore* for determining the correctness of a question is defined as follows:

$$MatchScore = \begin{cases} 1 & \text{if } X = True \text{ and } Y = True \\ 0 & \text{if } X \neq Y \\ J(A, B) & \text{if } X = False \text{ and } Y = False \end{cases} \quad (2)$$

Where  $X$  is the truth label of the participant’s answer,  $Y$  is the truth label of the correct answer,  $A$  is the set of actual false sentence, and  $B$  is the set of sentence marked by the participant. The correctness of the answers was determined using a equation (2), and an answer was considered correct if the *MatchScore* exceeded a threshold of 0.1. The fake news detection rates of the evaluators and GPT models were compared based on the proposed method, along with an evaluation of the subjective difficulty and domain knowledge.

*MatchScore* with a equation (2) was used to verify the participant answer. The average *MatchScore* for all participant questions



**Figure 2: Feature importance of information literacy estimation.**

was calculated, and participants were classified into high and low detection rate categories based on the median value. The features described in the proposed method were used for feature selection as shown in Table 1.

### 4.3 Information Literacy Classification

Participants are labeled as either high or low information literacy participants as explained in Section 4.2. A binary classification of high and low information literacy participants is applied using features explained in Table 1. Using a Support Vector Machine (SVM) facilitates the classification process. Five-fold cross-validation was conducted to evaluate the model’s generalization ability. This cross-validation excludes the same participants’ data inside the testing and training datasets. The approach ensured that the model performance did not depend on a specific participant.

## 5 RESULTS AND DISCUSSION

In this section, we state the results against two research questions.

Research question 1 is, “Can we classify high and low information literacy participants based on web browsing behaviors?” In order to collect high-low information literacy participants, we separated the participants’ datasets into two groups. Applying SVM towards the features shown in Table 1, we achieved a classification accuracy of 73.8 %. The result confirms that information literacy may be estimated from web browsing features.

Research question 2 is, “What characteristics can be observed when comparing participants with high and low information literacy?”. Figure 2 shows the frequency of selected features used for the binary classification. The result shows that the important web browsing features are search time ratio, average scroll speed, and time until the start of the search. Upon further examination of the data, *search\_time\_ratio* and *number\_of\_searches*, it was discernible that these values were notably higher in participants with higher skilled fake news identification ability. This finding suggests that the proportion of time and effort dedicated to searching, that is, the investment in searching, plays a significant role in estimating an individual’s ability to identify fake news. Moreover, analyzing the

*time\_from\_start\_to\_end* data revealed that participants with higher detection rates completed their answers in a shorter duration than those with lower rates. Interestingly, we also observed that *average\_scroll\_speed* for higher information literacy participants tend to scroll pages faster. From our analysis, it can be inferred that participants who demonstrate efficiency in processing and responding to information are more likely to be proficient in discerning fake news. This observation underscores the importance of information processing skills in fake news detection.

In summary, these insights collectively suggest that both the intensity of the search effort, as indicated by the search time ratio and the number of searches, and the efficiency in concluding the task, as shown by the time from start to end, are critical factors in estimating an individual’s ability to detect fake news. These findings highlight the nuanced relationship between search behavior and information verification skills in the context of fake news detection.

## 6 LIMITATION AND FUTURE WORK

This study compared high and low information literacy participants web behaviors. We found that the search time ratio, average scrolling speed, number of searches correlate with the high-low information literacy participants. However, it did not examine specific aspects, such as the types of sites visited or the textual evidence used in responses. Our future work direction is to evaluate the quality of searches and gain deeper insights.

Analyzing website types and page transition sequences could provide deeper insights into how search behaviors influence fake news detection. Furthermore, we will investigate whether intervention on supporting web browsing can enhance participants fake news detection capabilities. We aim to offer humans to make more effective strategy against fact-checking.

Finally, our participant pool for this study was largely homogeneous. A participant pool with a different make-up of gender, age, and nationality may result in different results. Future work should aim to include a diverse population in order to generalize the results of this work.

## 7 CONCLUSION

This study compares web browsing behaviors with high and low information literacy. Our study implements *Fake News Detective* and uses the open-source web browsing logger *TrackThink Camera* for data collection and collects data from 20 participants. Using SVM, we achieved the binary classification of high and low information literacy participants with an accuracy of 73.8 %. The feature importance shows that search time ratio is higher, number of searches are more, and average scroll speed was faster for high information literacy participants. Our findings show that web browsing features are factors in understanding the human skill of information literacy.

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